
Pooling Discards Resolution-Invariant Structure: An Architecture and Ablation Study on Darcy Flow

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Abstract

1 We study how architectural choices affect zero-shot resolution generalization in
2 neural operators using the standard 2D Darcy flow benchmark. We compare
3 CNN, FNO, U-FNO, UNO, and hypernetwork-conditioned UNO (HC-UNO),
4 together with no-pooling ablations of UNO and HC-UNO, training exclusively
5 at 16×16 resolution and evaluating zero-shot transfer at 32×32 . The main
6 results are: 1) FNO provides the strongest resolution robustness, with only a
7 $1.27\times$ increase in error under resolution shift, while the CNN degrades by $7.77\times$,
8 2) removing pooling substantially improves zero-shot robustness for both UNO
9 and HC-UNO, reducing their degradation factors from $1.88\times$ to $1.32\times$ and from
10 $2.12\times$ to $1.35\times$, respectively, and 3) hypernetwork-based input-conditioned filters
11 improve in-distribution accuracy but have little effect on resolution robustness,
12 which is primarily affected by the presence of pooling. These results indicate that
13 pooling is a key source of resolution sensitivity in U-shaped operator architectures,
14 while filter conditioning and resolution robustness can be treated as largely separate
15 architectural properties.

16 1 Introduction and related work

17 Neural operators seek to learn mappings between infinite-dimensional function spaces, with discretiza-
18 tion invariance being a key advantage of the framework. In principle, a well-posed operator learning
19 method should therefore generalize across grid resolutions, including resolutions not encountered
20 during training [1]. The Fourier Neural Operator (FNO) [2] was among the first architectures to
21 realize this property in practice by parameterizing spectral convolutions with a fixed, resolution-
22 independent set of Fourier modes. Follow-up work introduced U-shaped multiscale architectures,
23 including U-FNO [3] and U-NO [4], which combine spectral convolutions with a U-Net-style [5]
24 encoder-decoder hierarchy using spatial pooling and upsampling to increase representational capacity
25 and training efficiency.

26 Conditioning spectral filters on the input via a hypernetwork [6] has been explored outside the
27 operator-learning literature for vision and time-series token mixing. The Adaptive Frequency Filter
28 (AFF) token mixer [7], for example, generates frequency-domain filters dynamically for each input
29 instance, with the filters varying across inputs. Concurrently, within the operator-learning literature
30 itself, dynamic kernel generation via a hypernetwork has been used to replace FNO’s static spectral
31 kernel with an input-conditioned one [8], and instance-adaptive (rather than purely mode-adaptive)
32 spectral mixing has been studied in the Adaptive Fourier Neural Operator [9]. In this paper we
33 apply the same underlying idea, an input-conditioned filter generated by a hypernetwork, specifically
34 to UNO’s anisotropic Gaussian filter bank within a U-shaped hierarchy (HC-UNO), and study it
35 alongside a direct architectural ablation: removing the pooling operation from the U-shaped encoder-
36 decoder while keeping all other components fixed.

2 Method

2.1 Task and data

We consider the steady-state 2D Darcy flow equation, $-\nabla \cdot (a(x)\nabla u(x)) = f(x)$, on the unit square with Dirichlet boundary conditions, where $a(x)$ is a piecewise-constant diffusion coefficient field and $u(x)$ is the pressure field. We use the standard Darcy flow benchmark [2, 10]. We learn the solution operator $\mathcal{G} : a \mapsto u$ from data generated at a base resolution of 16×16 for training and evaluation, with a separate zero-shot test set at 32×32 generated from the same coefficient-field distribution and forcing term. Models are trained and selected exclusively on the 16×16 data. Training uses two NVIDIA T4 GPUs.

2.2 Architectures

Let $a \in \mathbb{R}^{1 \times H \times W}$ denote the diffusion coefficient field, $u \in \mathbb{R}^{1 \times H \times W}$ the target pressure field, and $\mathcal{N}_\Theta : a \mapsto \hat{u}$ the learned mapping. Define the generic spectral block

$$\mathcal{B}(\mathbf{h}; \mathcal{G}) = \sigma \left[\text{GN} \left(\mathcal{F}^{-1} \left[\sum_{j=1}^{N_f} \sum_{c=1}^{C_{\text{in}}} W_{j,c,\cdot} (G_j \odot \mathcal{F}(\mathbf{h})_c) \right] + W^{1 \times 1} \mathbf{h} \right) \right], \quad \sigma = \text{GELU}, \quad (1)$$

where $G_j \in \mathbb{R}^{H \times W_f}$ is a real, frequency-location-dependent window shared across channels, $W_j \in \mathbb{C}^{C_{\text{in}} \times C_{\text{out}}}$ is a complex channel-mixing matrix that does *not* depend on frequency location, and the sum over filters j and input channels c is carried out inside a single inverse transform.

CNN: The baseline uses no spectral component. Each block is

$$\mathcal{B}_{\text{CNN}}(\mathbf{h}) = \sigma \left[\text{GN} \left(\text{Conv}_{3 \times 3}^{(2)} \left(\sigma \left(\text{GN} \left(\text{Conv}_{3 \times 3}^{(1)}(\mathbf{h}) \right) \right) \right) \right) \right], \quad (2)$$

with no term analogous to $\mathcal{F}$, $\mathcal{F}^{-1}$, or G_j ; all mixing is spatially local.

FNO: The spectral operator uses a learned, mode-dependent Fourier multiplier instead of a separable filter-times-channel-matrix,

$$\mathcal{G}_{\text{FNO}} = \{R_l(k_x, k_y)\}_{|k| < \text{modes}}, \quad R_l(k_x, k_y) \in \mathbb{C}^{C_{\text{in}} \times C_{\text{out}}}, \quad (3)$$

$$\mathcal{B}_{\text{FNO}}(\mathbf{h}) = \sigma \left[\text{GN} \left(\mathcal{F}^{-1} [R_l \odot \mathcal{F}(\mathbf{h})] + W_l^{1 \times 1} \mathbf{h} \right) \right], \quad (4)$$

truncated to a low-mode window $|k| < \text{modes}$ and zero elsewhere, with independent weight blocks for the positive- and negative-frequency halves of the spectrum. FNO applies $\mathcal{B}_{\text{FNO}}$ at a single, fixed spatial resolution with no pooling or upsampling.

UNO: Each spectral filter is a real, channel-independent anisotropic Gaussian window defined directly in *absolute* frequency coordinates,

$$G_j(k_x, k_y) = \exp \left[- \left(\frac{u_j}{b_j} \right)^2 - \left(\frac{v_j}{b_j/3} \right)^2 \right], \quad \begin{pmatrix} u_j \\ v_j \end{pmatrix} = \begin{pmatrix} \cos \theta_j & \sin \theta_j \\ -\sin \theta_j & \cos \theta_j \end{pmatrix} \begin{pmatrix} k_x - c_{x,j} \\ k_y - c_{y,j} \end{pmatrix}, \quad (5)$$

with center frequencies $(c_{x,j}, c_{y,j}) \in [-F_{\text{max}}, F_{\text{max}}]^2$, bandwidth $b_j = \text{softplus}(\beta_j) + 0.5$, and orientation $\theta_j \in [0, \pi)$ all learned as *dataset-level* parameters shared across all samples, independent of the input a . Here F_{max} is fixed to the Nyquist frequency of the training resolution. $\mathcal{G}_{\text{UNO}} = \{G_j\}_{j=1}^{N_f}$ together with the channel-mixing weights $\{W_j\}_{j=1}^{N_f}$ of Eq. 1 are learned and shared across all samples and all hierarchy levels (after resizing).

UNO, U-FNO, and HC-UNO share the same encoder-decoder recursion,

$$\mathbf{s}^l = \mathcal{B}(\mathbf{h}^{l-1}; \mathcal{G}^l), \quad \mathbf{h}^l = \mathcal{P}(\mathbf{s}^l), \quad \mathbf{h}_{\text{dec}}^{l-1} = \mathcal{B}^l(\mathcal{U}(\mathbf{h}_{\text{dec}}^l) + \mathbf{s}^l; \mathcal{G}^l), \quad (6)$$

where $\mathcal{P}$ and $\mathcal{U}$ denote average pooling and bilinear upsampling, respectively, and $\mathcal{G}^l$ is the filter bank resized by bilinear interpolation to the spatial resolution at level l . Thus,

$$\boxed{\text{U-FNO} : \mathcal{B} = \mathcal{B}_{\text{FNO}}, \quad \text{UNO} : \mathcal{B} = \mathcal{B}(\cdot; \mathcal{G}_{\text{UNO}}).} \quad (7)$$

For the no-pool ablation of UNO and HC-UNO, $\mathcal{P}$ and $\mathcal{U}$ in Eq. 6 are replaced by the identity, so that all N_{blocks} spectral blocks are applied sequentially at the full input resolution,

$$\mathbf{h}^i = \mathcal{B}^i(\mathbf{h}^{i-1}; \mathcal{G}), \quad i = 1, \dots, N_{\text{blocks}}, \quad (8)$$

with the same total number of spectral blocks, channel width, and filter bank as the pooled variant, isolating pooling as the only architectural difference.

HC-UNO: The filter parameters are generated from the raw input by a hypernetwork,

$$\mathbf{q} = W_h \text{Flatten} \left[\text{Pool}_1 \left(\sigma \left(\text{Conv}_{3 \times 3}^{(2)} \left[\sigma \left(\text{Conv}_{3 \times 3}^{(1)} (\text{Pool}_8(a)) \right) \right] \right) \right) \right] \in \mathbb{R}^{N_f \times 4}, \quad (9)$$

$$(c_{x,j}, c_{y,j}, b_j, \theta_j) = (F_{\max} \tanh(q_{x,j}), F_{\max} \tanh(q_{y,j}), \text{softplus}(q_{b,j}) + 0.5, \pi \sigma(q_{\theta,j})), \mathcal{G}_{\text{HC}}(a) = \{G_j(a)\}_{j=1}^{N_f}. \quad (10)$$

The channel-mixing weights $\{W_j\}_{j=1}^{N_f}$ remain learned, dataset-level parameters, exactly as in UNO; only the filter geometry $\mathcal{G}_{\text{HC}}(a)$ is input-conditioned, computed once from the raw input a and reused, after resizing, at every level l . Hence the HC-UNO spectral block is $\mathcal{B}_{\text{HC}}(\mathbf{h}; a) = \mathcal{B}(\mathbf{h}; \mathcal{G}_{\text{HC}}(a))$, and the complete mapping is

$$a \xrightarrow{\text{Conv}_{1 \times 1}} \mathbf{h}^0 \xrightarrow[\mathcal{P}, \mathcal{U}, \text{skip}]{\mathcal{B}_{\text{HC}}(\cdot; a)} \mathbf{h}_{\text{dec}} \xrightarrow{\text{Conv}_{1 \times 1} \rightarrow \sigma \rightarrow \text{Conv}_{1 \times 1}} \hat{u}. \quad (11)$$

For HC-UNO (no-pool), Eq. 11 uses the block sequence of Eq. 8 in place of the pooled hierarchy, with $\mathcal{G}_{\text{HC}}(a)$ computed once and reused unresized at every block.

3 Results

Table 1: Relative L_2 error on in-distribution Darcy@16 and zero-shot Darcy@32, mean $\pm$ std over 5 seeds. Degradation is the ratio of zero-shot error to in-distribution error.

Architecture	Darcy@16 (test)	Darcy@32 (0-shot)	Degradation
CNN	0.0817 ± 0.0017	0.6345 ± 0.0561	$7.77\times$
FNO	0.1024 ± 0.0016	0.1297 ± 0.0008	$1.27\times$
U-FNO	0.1018 ± 0.0033	0.3501 ± 0.0376	$3.44\times$
UNO	0.1088 ± 0.0057	0.2047 ± 0.0149	$1.88\times$
HC-UNO	0.0991 ± 0.0029	0.2100 ± 0.0385	$2.12\times$
UNO (no-pool)	0.1034 ± 0.0083	0.1362 ± 0.0132	$1.32\times$
HC-UNO (no-pool)	0.0864 ± 0.0015	0.1165 ± 0.0022	$1.35\times$

Three patterns are visible in Table 1. First, the CNN baseline achieves the lowest in-distribution error of any architecture but collapses under resolution shift, degrading by a factor of $7.77\times$, far more than any spectral or spectral-hybrid architecture. This is consistent with the CNN’s lack of any discretization-invariant component: its convolutional kernels and any resolution-dependent layers are tied to the 16×16 grid on which they were trained.

Second, removing pooling improves both in-distribution accuracy and zero-shot robustness for both UNO and HC-UNO. UNO’s degradation factor falls from $1.88\times$ to $1.32\times$ without pooling, while its test error remains essentially unchanged. For HC-UNO, the degradation factor decreases from $2.12\times$ to $1.35\times$, while the test error improves from 0.0991 to 0.0864 , the best in-distribution result among the architectures tested other than the CNN. HC-UNO (no-pool) also achieves the lowest zero-shot error among the U-shaped variants. These results indicate that pooling can introduce a resolution-dependent effect in U-shaped operator architectures, despite its usual role in increasing receptive fields and reducing computational cost. In this setting, removing pooling substantially narrows the zero-shot robustness gap between UNO/HC-UNO and FNO.

Third, FNO remains the most resolution-robust architecture in absolute terms, with a degradation factor of only $1.27\times$, marginally better than either no-pool variant. Its spectral convolution operates on a fixed set of Fourier modes without an explicit spatial downsampling step. U-FNO, which combines FNO-style spectral blocks with a pooled U-shaped hierarchy, exhibits the second-largest degradation at $3.44\times$. Together with the no-pool results above, this suggests that pooling contributes substantially to the resolution sensitivity observed in the U-shaped architectures.

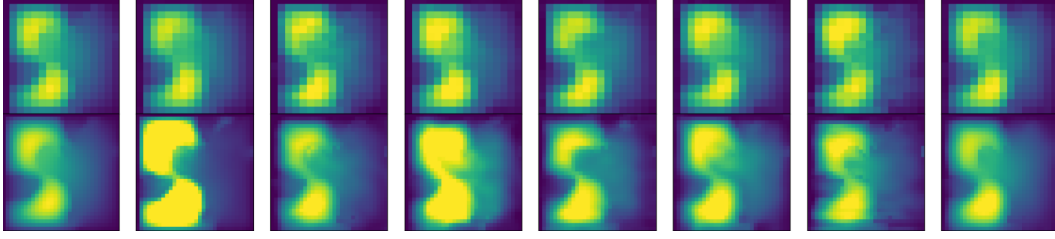


Figure 1: Qualitative comparison of Darcy flow predictions across model architectures. The top row shows the solution u at the training resolution of 16×16 , while the bottom row shows the corresponding zero-shot predictions at 32×32 . From left to right, the columns show the ground-truth solution, CNN, FNO, U-FNO, UNO, HC-UNO, UNO (no-pool), and HC-UNO (no-pool).

4 Discussion and Conclusion

Across the architectures and pooling ablations, spatial downsampling is associated with reduced zero-shot resolution transfer, regardless of the spectral parameterization. We hypothesize that average pooling introduces a scale assumption tied to the training resolution, such as a fixed receptive-field-to-grid-size ratio across hierarchy levels. Removing pooling eliminates this assumption, improving resolution robustness at the cost of higher memory usage while largely preserving the in-distribution accuracy benefits of the U-shaped architecture and, for HC-UNO, input-conditioned filters.

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